UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

BIOLOGY **5090/01**

Paper 1 Multiple Choice **October/November 2007**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

* 3 3 0 4 8 8 1 3 2 3 *

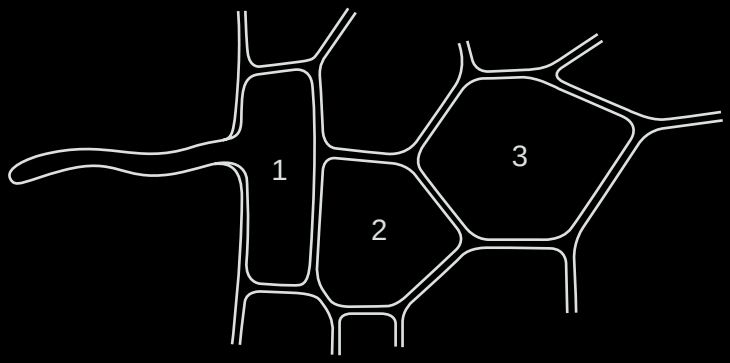
READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.



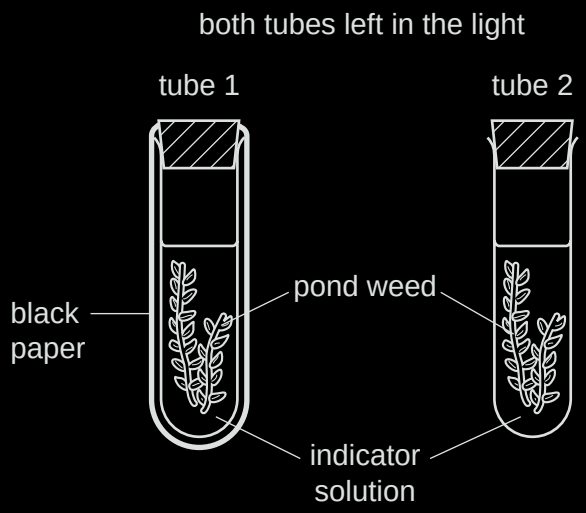
How does the water potential of the cell marked 2 differ from the water potential of the cell marked 3 and how do mineral ions mainly enter the cell marked 1?

	water potential of the cell marked 2	mineral ions mainly enter the cell marked 1
A	higher than cell 3	by active transport
B	higher than cell 3	by diffusion
C	lower than cell 3	by active transport
D	lower than cell 3	by diffusion

3 The mass of a cube of fresh potato is found. It is then placed in a test-tube containing a dilute solution of sucrose. After a day, its mass has increased.

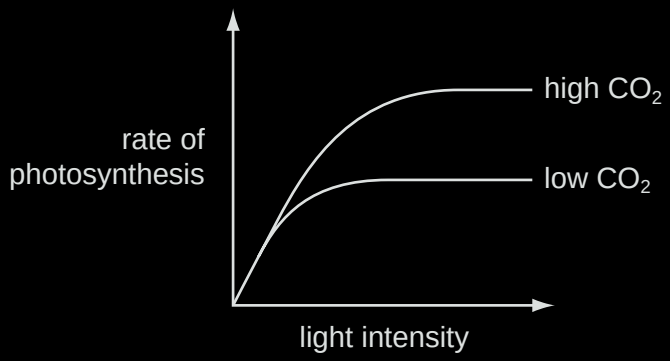
Which process has occurred and what has happened to the concentration of the sucrose in the solution in the test-tube?

	process	sucrose concentration
A	active transport	decreased
B	active transport	increased
C	osmosis	decreased
D	osmosis	increased



Which colours would the indicators show after three hours?

	tube 1	tube 2
A	purple	orange
B	orange	yellow
C	yellow	purple
D	yellow	orange



Which statement is supported **only** by evidence from the graph?

A Carbon dioxide limits the rate of photosynthesis at low light intensities.

B Light intensity and carbon dioxide both limit the rate of photosynthesis.

C The rate of photosynthesis is proportional to light intensity.

D Temperature affects the rate of photosynthesis.

8 A person tries eating a diet consisting only of lettuce leaves and water.

Which condition might develop?

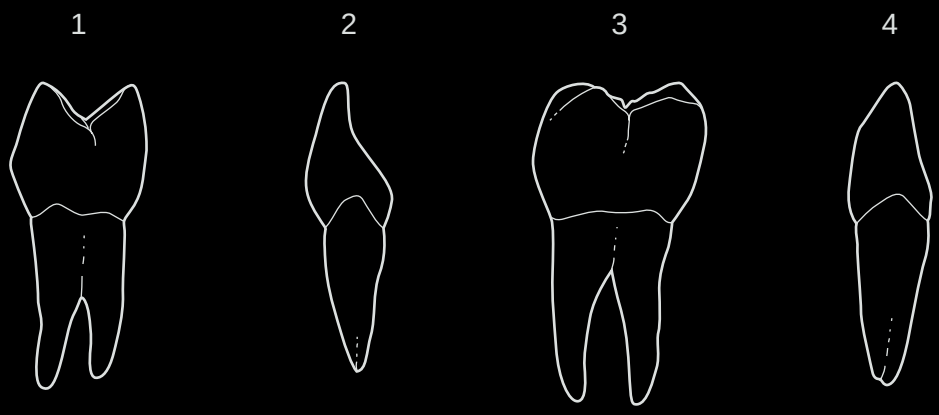
A constipation

B heart disease

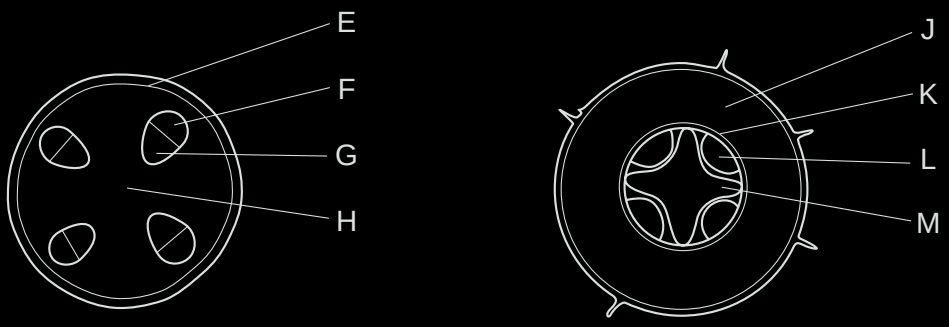
C rickets

D scurvy

9 Which of these human teeth are used for grinding?



A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

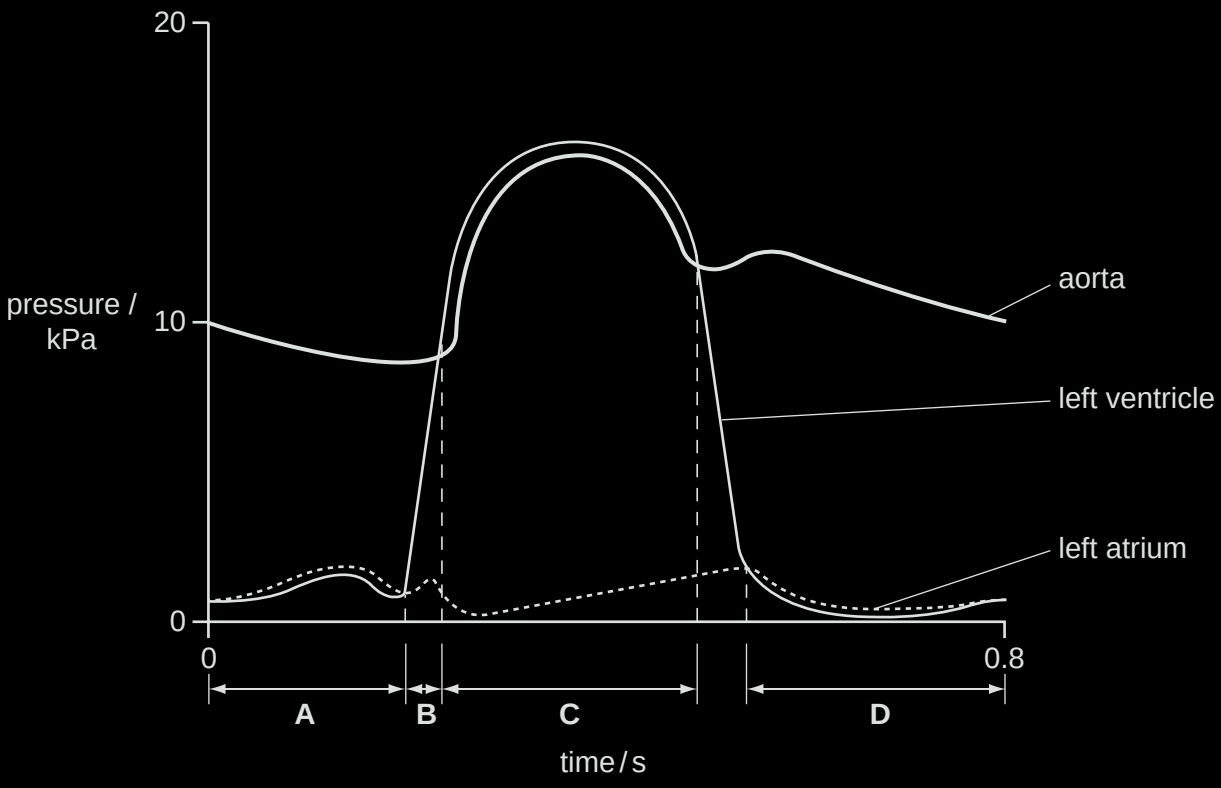


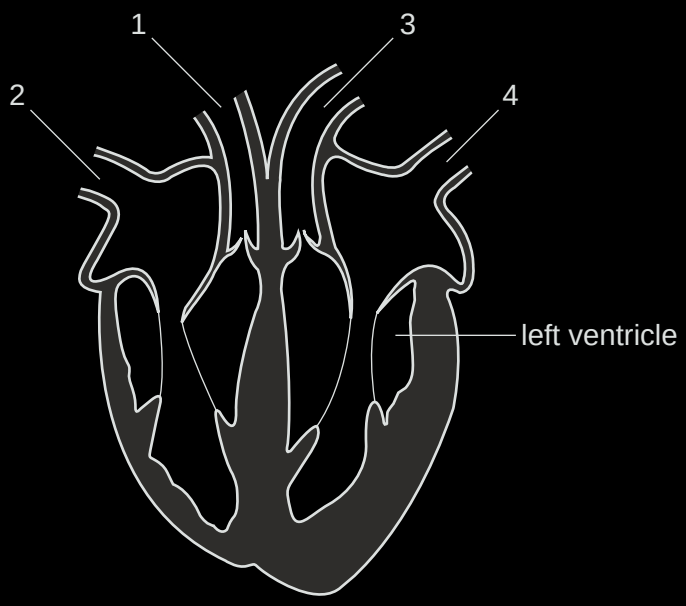
Which structures transport sugars through the stem and root?

A E and J **B** F and L **C** G and M **D** H and K

12 Which adaptations of a root hair cell make it suitable for water uptake?

	partially permeable cell membrane	surface area to volume ratio of the cell
A	absent	high
B	absent	low
C	present	high
D	present	low


20
10
0
pressure / kPa
aorta
left ventricle
left atrium
0
0.8
A
B
C
D
time / s



What are the functions of the numbered blood vessels?

	carries blood to body	carries blood to lungs	carries blood from lungs	carries blood from body
A	1	2	3	4
B	1	3	4	2
C	2	4	3	1
D	3	1	4	2

15 After muscular exercise, which blood vessel carries the **lowest** concentration of carbon dioxide?

A hepatic artery

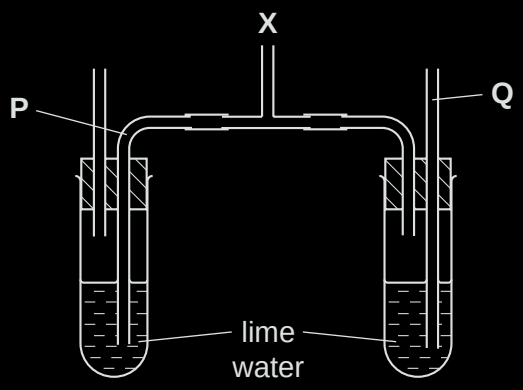
B pulmonary vein

C renal artery

D vena cava

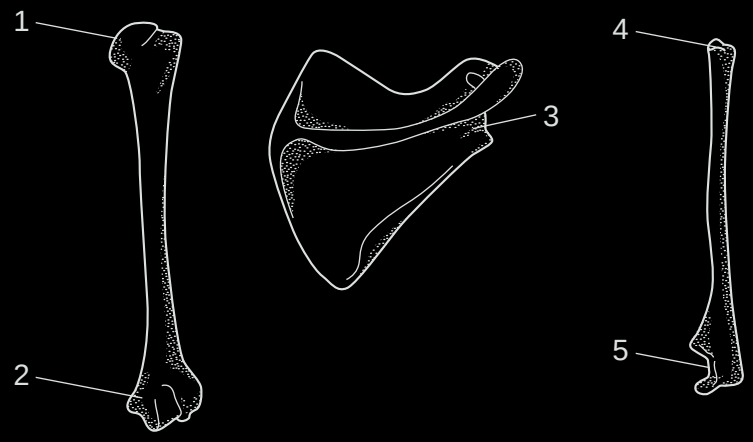
16 Which changes occur as a person breathes in deeply?

	diaphragm muscle	external intercostal muscles
A	contracts	contract
B	contracts	no change
C	relaxes	contract
D	relaxes	relax



What are the carbon dioxide concentrations at **P** and **Q**?

	CO_2 at **P** (%)	CO_2 at **Q** (%)
A	16	20
B	0.04	4.00
C	4.00	0.04
D	20	16



Between which parts is a ball and socket joint formed?

A 1 and 3 **B** 1 and 5 **C** 2 and 5 **D** 3 and 4

20 Urea is removed by the kidneys.

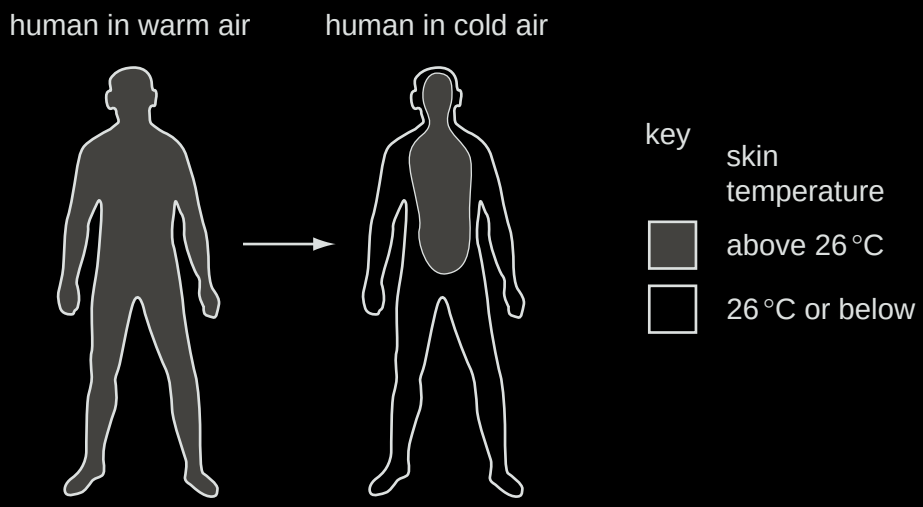
What is this process called?

A dialysis

B diffusion

C egestion

D excretion



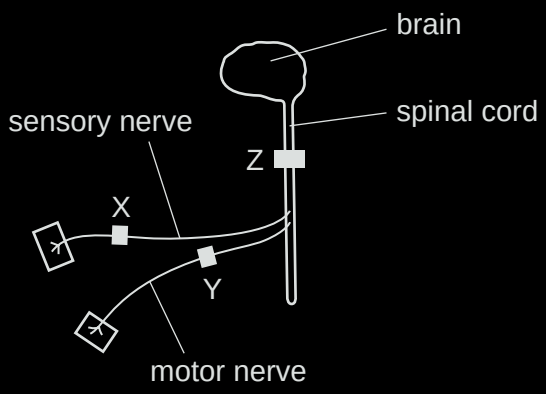
What causes the observed change in skin temperature on exposure to cold air?

A less blood flowing just below the skin

B less blood going to the heart and lungs

C more blood flowing just below the skin

D more blood going to the heart and lungs



Of four men, one had no anaesthetic block and the other three had only one anaesthetic block at X, Y or Z.

One of the men can move his leg in response to a pinprick, but does not feel it.

Where is the anaesthetic block?

A block is at X

B block is at Y

C block is at Z

D no block

23 Which is a function of adrenalin?

A to convert glycogen into glucose

B to decrease the breathing rate

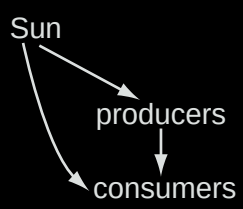
C to increase the rate of peristalsis in the ileum

D to stimulate cells of the liver to take up glucose

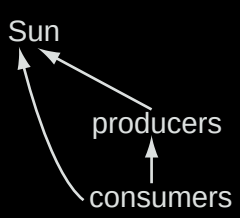
24 What happens to these structures when focussing on an object near the eye?

	ciliary muscles	suspensory ligaments
A	contract	tighter
B	contract	looser
C	relax	tighter
D	relax	looser

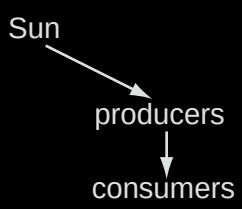
A


Sun
producers
consumers

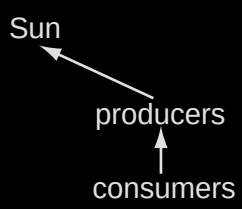
B

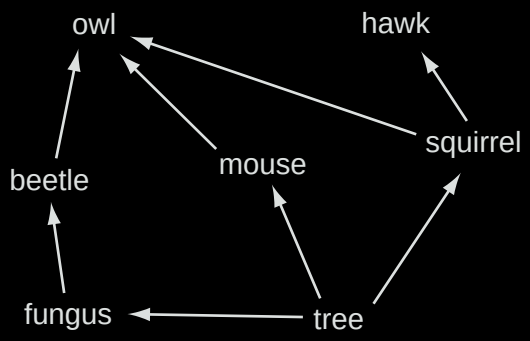

Sun
producers
consumers

C


Sun
producers
consumers

D


Sun
producers
consumers



Which of the organisms, shown in the food web, can survive by taking in only simple inorganic materials?

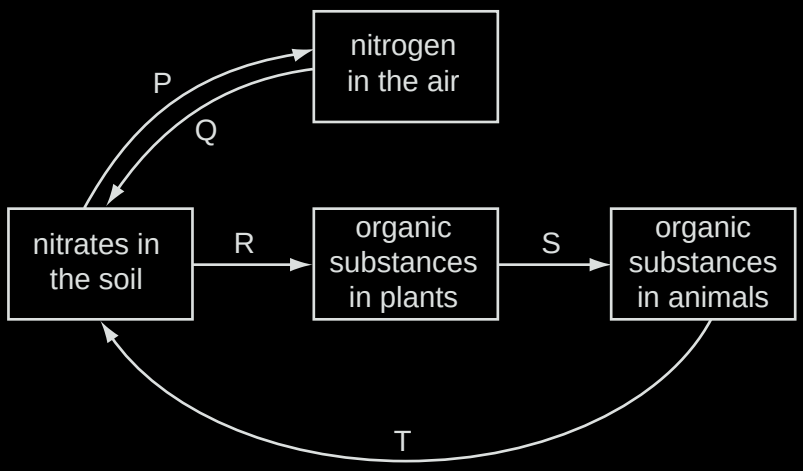
A beetle

B fungus

C owl

D tree

30 The diagram shows part of the nitrogen cycle.



Which stages involve bacteria?

A P, Q and T **B** P, R and S **C** Q, S and T **D** R, S and T

31 Some samples are taken from a human patient.

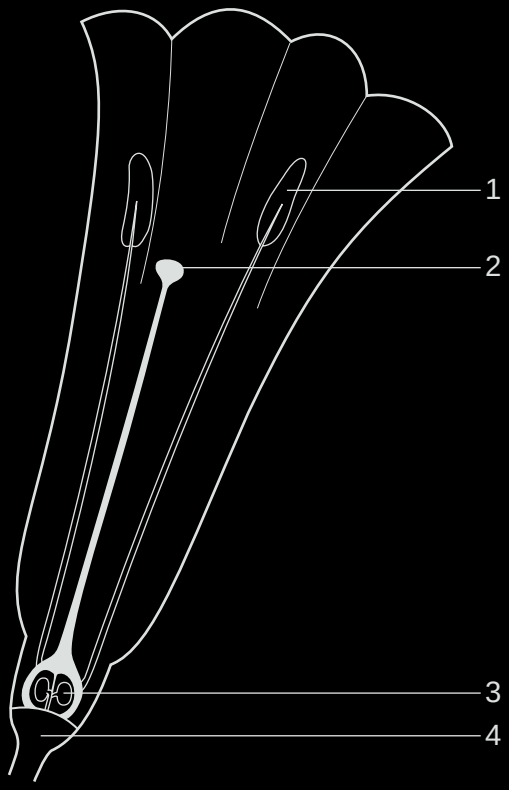
Which sample could be examined to find out whether the patient is infected with the malarial parasite?

A red blood cells

B saliva

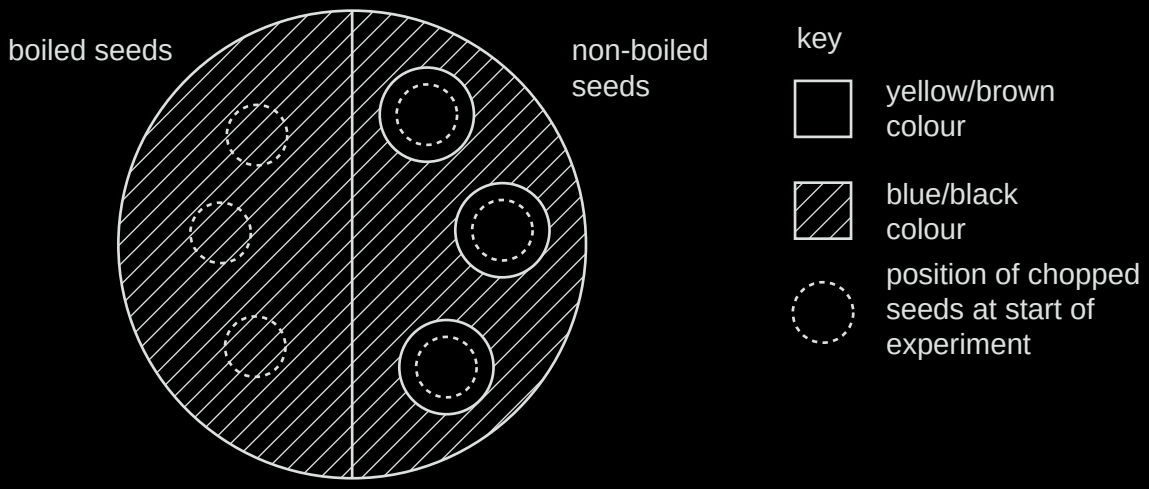
C urine

D white blood cells



In which structures are haploid nuclei formed by reduction division?

A 1 and 4 **B** 2 and 3 **C** 1 and 3 **D** 2 and 4



What is the explanation for the results with the non-boiled bean seeds?

A They absorb iodine.

B They absorb starch.

C They contain acid.

D They contain amylase.

35 Which method of contraception is the most effective in preventing the spread of diseases such as syphilis and HIV / AIDS?

A cervical caps / diaphragms

B male condoms

C oral contraceptive pills

D spermicides

36 What would be the result of cutting the sperm ducts on the right and left sides in a man?

A He would become sterile.

B He would be unable to develop sperms.

C He would be unable to pass urine.

D Male sex hormones would no longer circulate in the blood.